

Hyperloop Development Program

Reporting Period 3: Jan 2022–Jun 2022

31-08-2022 – V1

This report highlights the progress of the Hyperloop Development Program. First section summarizes the highlights within the growing hyperloop market, the second section provides an overall program overview, followed by section of progress per working group, and summarized in the financial overview.

HIGHLIGHTS WITHIN HYPERLOOP MARKET

The past months have shown highlights in the Global developments in the Hyperloop Market:

- **Feb:** Hardt and HDP announce to start Hyperconnected Europe project: [Link](#)
- **Feb:** Zeleros will roll out its hyperloop technology in the Port of Sagunto to demonstrate its container transport system: [Link](#)
- **Feb:** Virgin Hyperloop to refocus its efforts to cargo transport: [Link](#)
- **Mar:** Nevomo has completed the first phase of Magrail track construction: [Link](#)
- **Mar:** Venice – Padua: Italy's Ministry of Infrastructure has signed a memorandum of understanding with the region of Veneto and the regional highway operator CAV to begin a multistage process of developing a hyperloop system between the cities of Venice and Padua: [Link](#)
- **May:** Meet-up in Belgium organized by VIL for possible strategic partner in rollout Hyperloop network: [Link](#)
- **May:** Hardt and RFI extend cooperation on hyperloop study in Italy: [Link](#)
- **June:** Curiosity Inc. stream features hyperloop in its "Engineering the Future" program: [Link](#)
- **June:** €17.5M investment in Polish Nevomo to upgrade existing rail to hyperloop tech: [Link](#)
- European Commission has included hyperloop in its Smart Mobility Strategy and it is included in their proposal of the new transport policy (TEN-T). Further to be decided into Q3/Q4 2022.

HDP published 4 communication moments based on deliverables of the Program in the past half year:

- **Feb:** EHC to build temporary Phase A in Veendam
- **Apr:** EHC Fieldworks investigations started
- **Jun:** Results Hyperconnected Europe
- **Jun:** EHC steel production started Tata Steel & POSCO

HDP has onboarded INTIS to the program as Work Package Partner, to study Hyperloop research and demonstration in their Transrapid test facility in Lathen.

HDP has appointed Coen de Ronde as Director on behalf of the Stichting HDP.

HDP has focused on three means to raise additional funding:

- *Nationaal Groeifonds:* Over the past months Stichting HDP has worked intensively with Ministry IenW to prepare a proposal for round 3 of the Nationaal Groeifonds.

REPORTING ON OVERALL PROGRAM PROGRESS

This report is to confirm costs incurred and progress in the Hyperloop Development Program over the period Jan 2022-Jun 2022. The key program deliverable and key deliverable for this stage per working group are represented below.

WG	Key Program deliverable	Key deliverable Stage 3 (Summer 22 – Summer 23)
WG Cargo	Demonstrated hyperloop for cargo system (TRL-7) in the EHC	Detail Design Finished
WG EHC	European Hyperloop Center (EHC) Phase A completed	EHC Delivery and start first tests
WG FP	Identify and develop 5 routes up to a preliminary level, of which 1 with a go-ahead build-out of route	3 Preliminary route studies concluded
WG Passengers	Passenger hyperloop system concept (TRL-3 & SRL-5)	System Concept Design Definition finished
WG Program Management	Establish a connected hyperloop ecosystem	Plan for future funding of the ecosystem, finished spatial planning workshop

HDP has progressed into Stage 3 (June 22 – Summer 23) of a total of four stages. The progress of the Working Groups and Work Packages is presented in figure 1. Green means the Work Package is delivered, Yellow the Work Package is on-going.

The EHC Working Group is progressing on the temporary project location in Veendam, and the first partners and suppliers have been contracted for the construction. The Consortium Agreement is signed by all partners. The overall HDP program expenditure is delayed compared to planned, due to delay within WG EHC, which can be seen in the financial section of this report.

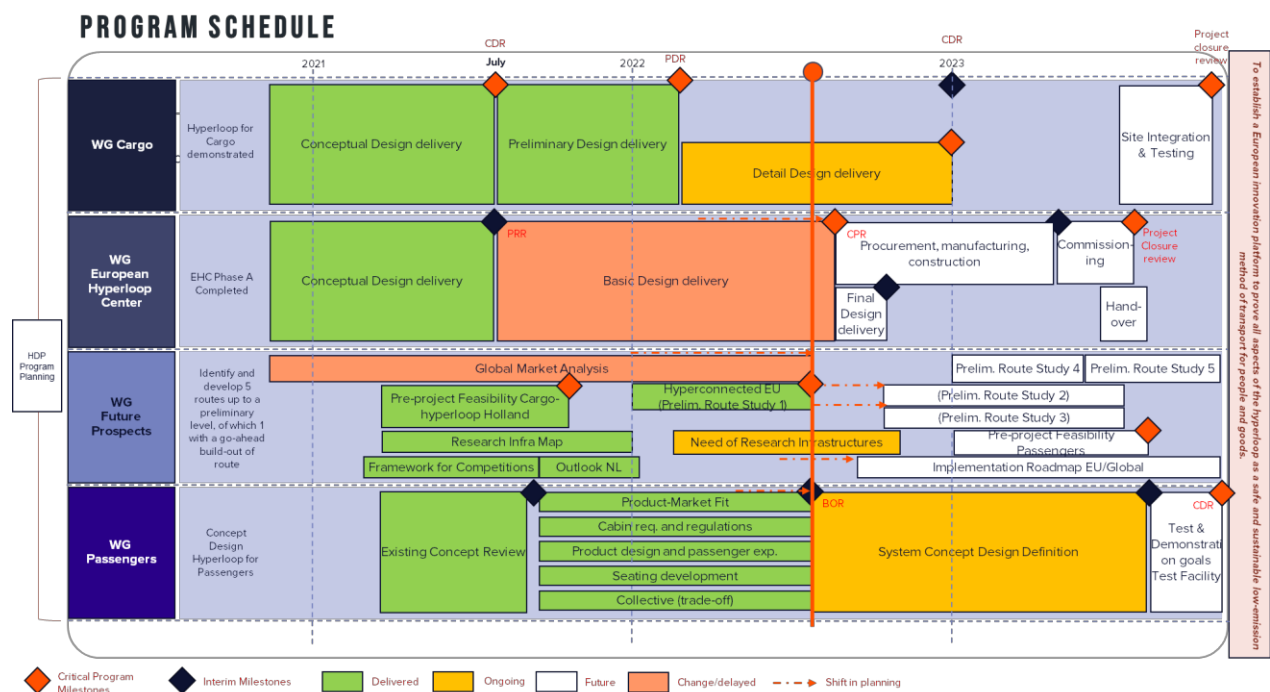


Figure 1: Stages & Milestone planning HDP

The detailed overview of program deliverable status for open stage 1, stage 2, and started stage 3 is presented below. A number of milestones from stage 1 and 2 have encountered delays, and will be completed in Q3 of 2022.

#	Milestone	Working Group	Status	(Adj.) Date delivery
Open from Stage 1:				
1.9	Final Market opportunity attractiveness	Future Prospects	Delayed	30/8/2022
1.16	Consortium Agreement HDP signed	Program Management	Finished	Q1-22
Stage 2:				
2.1	Basic Design EHC Approved	EHC	On-going (Delayed)	01/09/2022
2.2	Critical Funding and resources are secured for EHC.	Program Management + EHC	On-going	01/09/2022
2.3	EHC Permits approved (Basic Design Approved)	EHC	On-going (Delayed)	01/09/2022
2.4	Partners/suppliers contracted for Manufacturing & Assembly.	EHC	On-going (Delayed)	01/10/2022
2.5	Preliminary Design Cargo approved	Cargo	Finished	01/02/2022
2.6	Pre-project feasibility Cargo finished	Future Prospects	Finished	01/07/2021
2.7	Business Opportunity Definition completed	Passengers	Finished	30/04/2022
2.8	Scheme of operations for Hyperloop Research Infrastructure	Future Prospects	On-going (Delayed)	30/09/2022
Ongoing Stage 3:				
3.1	Need for Research Infrastructure completed	Future Prospects	On-going	30/09/2022
3.4	Preliminary Route #1: Hyperconnected Europe	Future Prospects	Finished	01/06/2022
3.5	Preliminary Route #2	Future Prospects	Not Started	01/12/2022
3.6	Preliminary Route #3	Future Prospects	Not Started	01/12/2022
3.7	Detail Design Cargo Approved	Cargo	On-going	31/12/2022
3.9	EHC Delivery & Start first tests	EHC	Not Started	Q2/Q3 2023
3.10	Project hand-over EHC	EHC	Not Started	Q2/Q3 2023
3.11	Roadmap EU Finished	Future Prospects	Not Started	Q2/Q3 2023
3.12	Work Package Research Infra finished	Future Prospects	Not Started	Q2/Q3 2023

In 2022-H1 five expert sessions have been organized on the research themes of the HDP. The purpose of the sessions is to review work, share implications and insights about the work being done in the HDP, with the aim of knowledge sharing and transfer.

- Technology as review of preliminary design Cargo (February)
- Standards and Safety: on Certification Roadmap (February)
- Public Adoption on passenger WG and as part of the Preliminary Route Study #1 (March)
- Socio-Cost Benefit as part of the Preliminary route study #1 (May)
- Integration as part of the Preliminary route study #1 (May)

In 2022-H2 four expert sessions will be organized on the research themes of the HDP:

- Ecosystem on work need for Hyperloop Research Infra (Q3)
- Standards and Safety
- Technology as review of the detail design Cargo (Q4)
- Socio-cost benefit and Integration as part of the Spatial Planning project (Q3/Q4)

REPORTING PER WORKING GROUP

WG Cargo

Status: Proceeding according to WG Plan

The Working Group Cargo finished reporting of the Preliminary Design Phase in Q1, with an extensive review week with partners as well as external experts. The WG progresses with the Detail Design Phase (to be delivered Q4-22/Q1-23):

- POSCO and TATA Steel have started with the production of the steel for the European Hyperloop Center Phase A. Furthermore both are working on numerous research projects for, among others, the tube materials and shapes.
- Mercon is working on the attachments of the internals to the tube.
- Denys has started on the research project around hyperloop tunnels.
- Hardt is working on the vehicle design, infrastructure and the tractive work packages.

Standardization, regulations

On June 3rd in the quarterly meeting between the EU DG Move, S2R Joint undertaking and hyperloop developers, DG Move presented a new timeline for the EU regulatory development for hyperloop. The aim is adoption of a basic hyperloop EU Regulatory Framework in 2025 by European Council and European Parliament. The focus is primarily passenger transport with the possibility to define a few requirements for cargo transport. The next major step is the launch of the impact assessment with the expected completion in the first half of 2023. This impact assessment for hyperloop should define the initial level of input and requirements on safety, security, environmental requirements (such as noise, land use, CO2 emissions) as well as interoperability.

The working group has finished two main deliverables:

- [HDP-WG-CARGO-2.5]- WG Cargo - Preliminary Design Phase Summary

Next key milestone:

- [HDP-WG-Cargo-3.8] - Detail Design Summary Document Cargo (Q4 2022)

WG Passengers

Status: Proceeding according to Schedule

The Working Group Passenger (NS, Recaro, Schweizer, Hardt, ADSE) has finished the *Business opportunity definition*, where three design scenarios, Small, Medium, Large, are investigated and trade-off what are the design implications for the passenger when choosing one of the scenarios. Preliminary conclusions are to move forward with the medium scenario. The results have also been presented in the expert session as part of the Preliminary Route study #1.



The team has finished main deliverables:

- [HDP-WG-Pax-2.7] - Delivery Business Opportunity Definition (March 2022)

Next key milestones:

- Research on Passenger travel perception (Q3)
- [HDP-WG-Pax-4.5] - Conceptual Design Document Passenger (H2-2023)

WG EHC

Status: Change to baseline

In the past half year it has been decided to temporary build up the Phase A on a different location in the Municipality of Veendam, to maintain the schedule time to open the facility. Since then, the Working Group has started with field works to investigate the soil conditions, and have started with design of the required substructure and foundations, as well as the required permit procedures. Partners TATA Steel and POSCO have started production of the steel tubes, to be produced as pipes in Q3 for the EHC, marking the start of the realization phase of the project. The updated schedule defines on-site activities by Q4-22 and into 2023, with opening date Q2/Q3 2023.



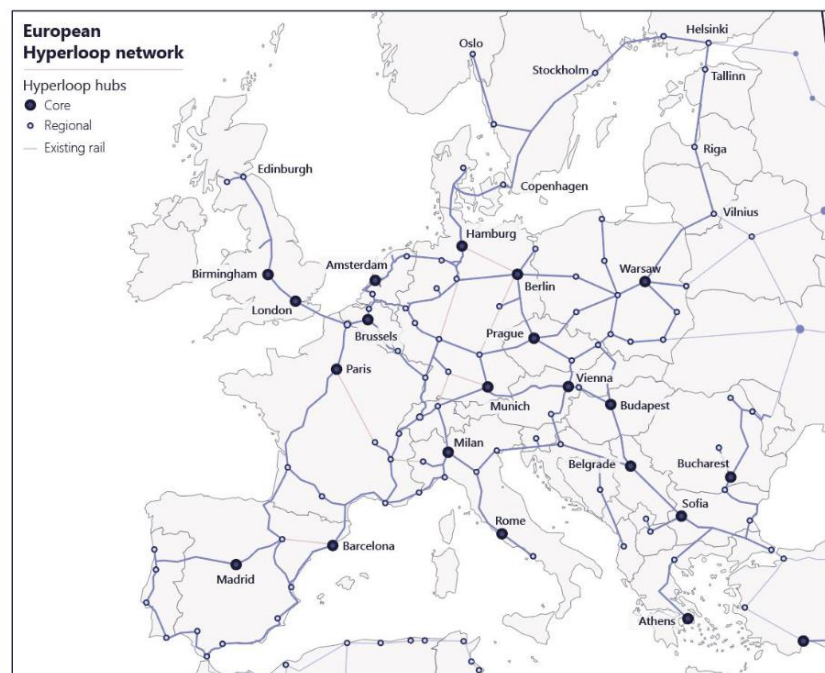
Intended location HUSA site in Veendam

WG Future Prospects

Status: Proceeding according to WG Plan

The focus of the Working Group was targeted on the preliminary route study #1, the “HyperConnected EU” initiative, and on the need for research infrastructure.

The HyperConnected EU results show that a 25,000-kilometer network could connect 130 cities covering mainland Europe with transport times comparable to aviation. It would require a total estimated investment of 981 billion Euro – about 10% of the total investment need in transport infrastructure in Europe towards 2050. The investment could be recouped with passenger fares and transport rates similar to aviation and trucking while reducing the overall investment need in other modalities. Hyperloop helps achieve Green Deal goals of net zero emissions by potentially reducing CO2 emissions with 113 to 242 million tonnes p.a. by 2050. The Hyperconnected Europe initiative is officially supported by Berlin, Munich (Germany), Province of Groningen, Province of North Holland and Rotterdam (Netherlands), Flanders (Belgium), Brasov (Romania), Rzeszow (Poland) as well Zaragoza and Algeciras (Spain).



The European Hyperloop Network

The WG has finished main deliverables:

- [HDP-WG-FP-3.4] – Hyperconnected Europe (Q2)
- Interactive Network Tool
- A Hyperloop Handbook for Public and Private Stakeholders
- On the Future Development of Hyperloop Infrastructure

Next key milestones:

- Need for future hyperloop research infrastructure
- Preliminary route study 2 & 3.
- Annual hyperloop outlook: the Netherlands